24ADI003_EXPERIMENT_4

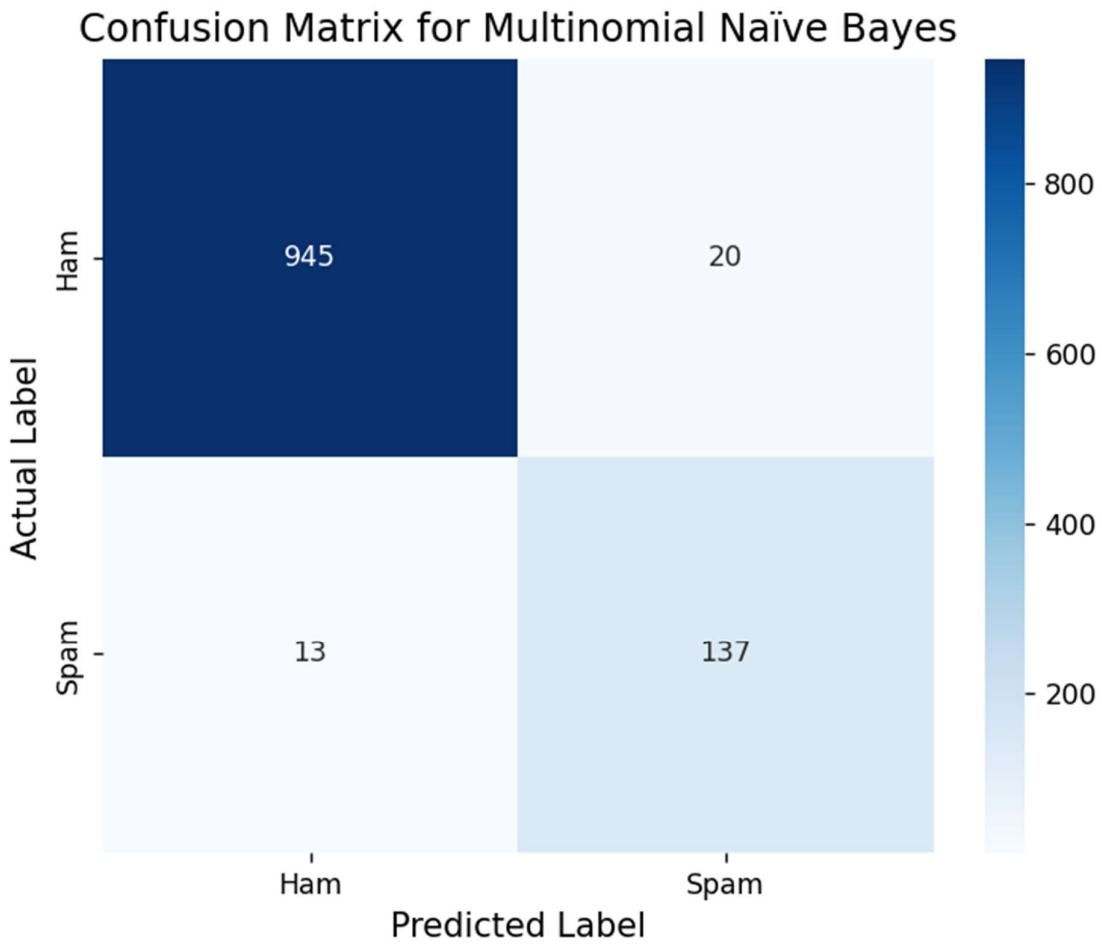
1.Scenario 1:Gaussian Naïve Bayes

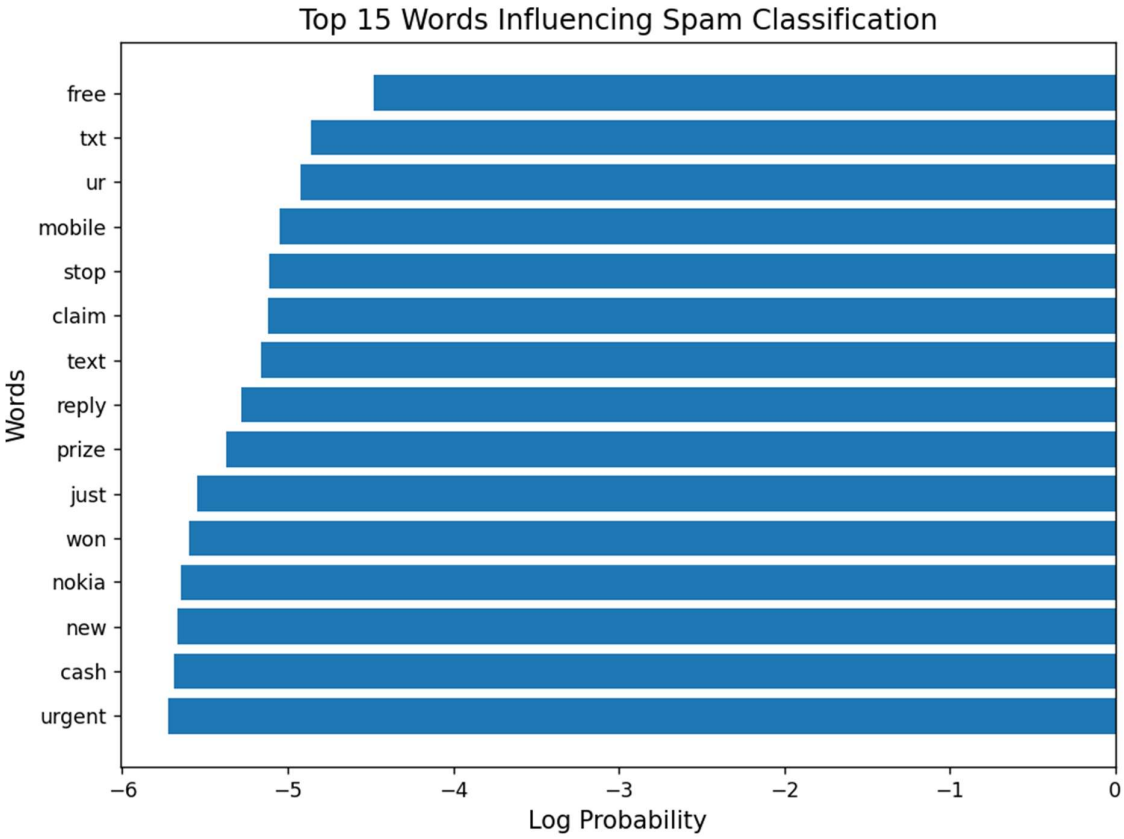
Output:

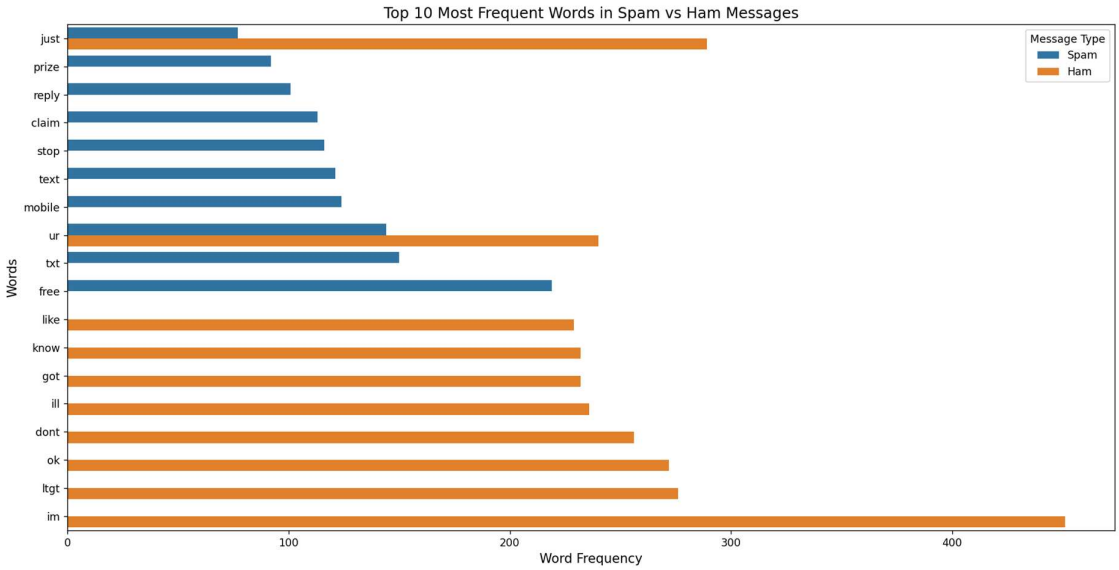
```
JANASREE 24BAD040
Accuracy: 0.9704035874439462
Precision: 0.8726114649681529
Recall: 0.9133333333333333
F1 Score: 0.8925081433224755
              precision    recall  f1-score   support

           0       0.99      0.98      0.98       965
           1       0.87      0.91      0.89       150

    accuracy                           0.97      1115
   macro avg       0.93      0.95      0.94      1115
weighted avg       0.97      0.97      0.97      1115

Message: go until jurong point crazy available only in bugis n great world la e buffet cine there got amore wat
Actual: ham
Predicted: spam
------
Message: k tell me anything about you
Actual: spam
Predicted: ham
------
Message: k did you call me just now ah
Actual: ham
Predicted: spam
------
Message: for real when u getting on yo i only need  more tickets and one more jacket and im done i already used all my multis
Actual: ham
Predicted: spam
```

```
------
Message: okay name ur price as long as its legal wen can i pick them up y u ave x ams xx
Actual: ham
Predicted: spam
------
Accuracy with smoothing: 0.9659192825112107
>
```

Figures:


Confusion Matrix for Multinomial Naïve Bayes
Actual Label
Ham
Spam
945
20
13
137
Ham
Spam
Predicted Label
800
600
400
200


Top 15 Words Influencing Spam Classification
Words
free
txt
ur
mobile
stop
claim
text
reply
prize
just
won
nokia
new
cash
urgent
−6
−5
−4
−3
−2
−1
0
Log Probability


Top 10 Most Frequent Words in Spam vs Ham Messages
Message Type
Spam
Ham
Words
just
prize
reply
claim
stop
text
mobile
ur
txt
free
like
know
got
ill
dont
ok
ltgt
im
0
100
200
300
400
Word Frequency

2.Scenario 2: Gaussian Naïve Bayes

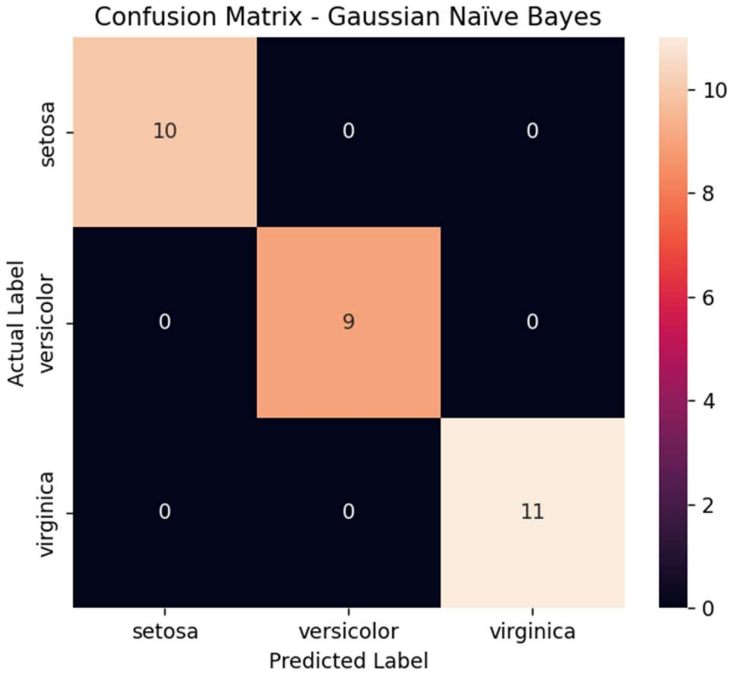
Output:

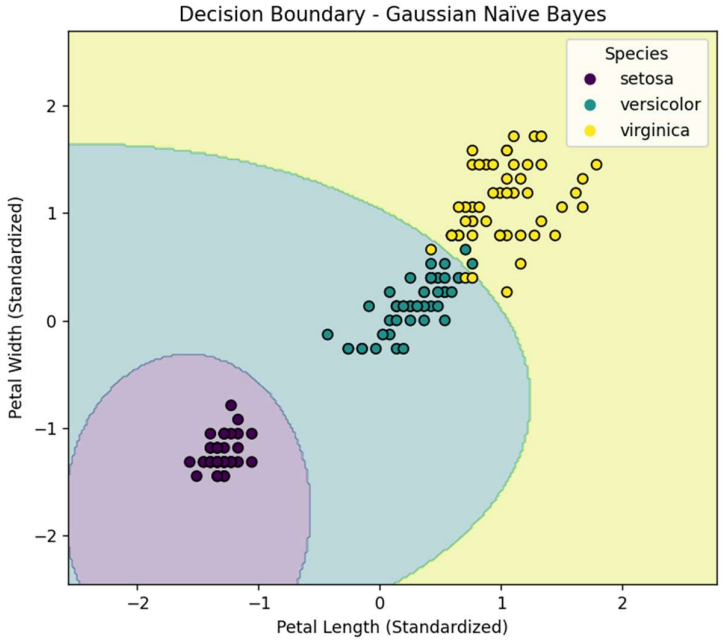
```
JANASREE 24BAD040
IRIS CLASSIFICATION USING GAUSSIAN NAÏVE BAYES
   sepal length (cm)  sepal width (cm)  ...  petal width (cm)  species
0                5.1               3.5  ...               0.2        0
1                4.9               3.0  ...               0.2        0
2                4.7               3.2  ...               0.2        0
3                4.6               3.1  ...               0.2        0
4                5.0               3.6  ...               0.2        0

[5 rows x 5 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
None
       sepal length (cm)  sepal width (cm)  ...  petal width (cm)     species
count         150.000000        150.000000  ...        150.000000  150.000000
mean            5.843333          3.057333  ...          1.199333    1.000000
std             0.828066          0.435866  ...          0.762238    0.819232
min             4.300000          2.000000  ...          0.100000    0.000000
```

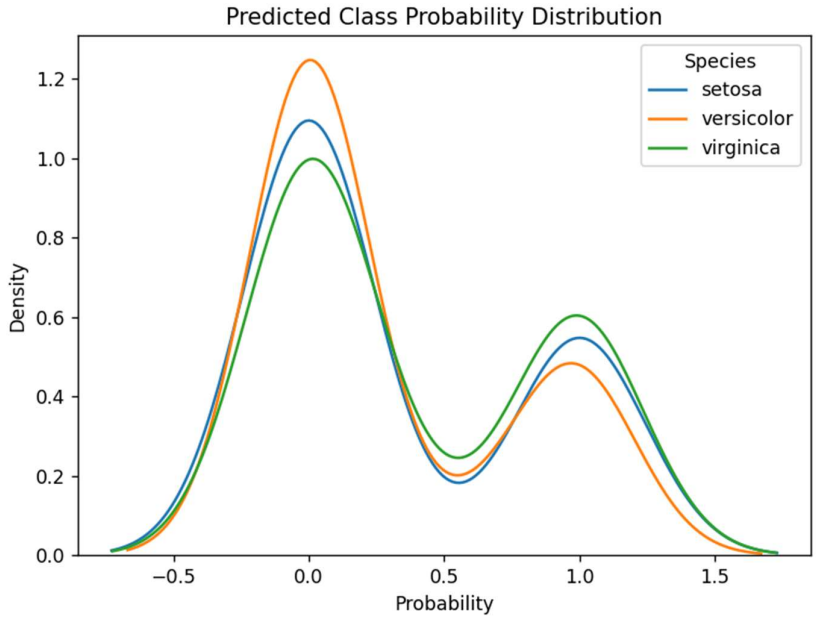
```
25%        5.100000    2.800000  ...    0.300000    0.000000
50%        5.800000    3.000000  ...    1.300000    1.000000
75%        6.400000    3.300000  ...    1.800000    2.000000
max        7.900000    4.400000  ...    2.500000    2.000000

[8 rows x 5 columns]
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
species              0
dtype: int64
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

```
      Actual   Predicted
0  versicolor  versicolor
1      setosa      setosa
2   virginica   virginica
3  versicolor  versicolor
4  versicolor  versicolor
5      setosa      setosa
6  versicolor  versicolor
7   virginica   virginica
8  versicolor  versicolor
9  versicolor  versicolor
         setosa    versicolor     virginica
0   5.973226e-90  9.956358e-01  4.364232e-03
1   1.000000e+00  4.961577e-14  6.549219e-21
2  7.318611e-290  4.929473e-12  1.000000e+00
3   2.818383e-94  9.775936e-01  2.240644e-02
4  1.138764e-105  8.700226e-01  1.299774e-01
Logistic Regression Accuracy: 1.0
```

Figures:


Confusion Matrix - Gaussian Naïve Bayes
Actual Label
setosa
versicolor
virginica
10
0
0
0
9
0
0
0
11
setosa
versicolor
virginica
Predicted Label
10
8
6
4
2
0


Decision Boundary - Gaussian Naïve Bayes
Species
setosa
versicolor
virginica
Petal Width (Standardized)
Petal Length (Standardized)
2
1
0
−1
−2
−2
−1
0
1
2


Predicted Class Probability Distribution
Species
setosa
versicolor
virginica
Density
Probability
0.0
0.2
0.4
0.6
0.8
1.0
1.2
−0.5
0.0
0.5
1.0
1.5